



INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR

ME 333: Mechanics of Materials

Charpy V-notch Test

Group 9

Atharva Prasad Suryarao	23110051
Mukesh Dewangan	23110211
Vivek Kumar	23110361

Professor :

Prof. Ravi Sastri Ayyagari
Prof. Harmeet Singh

August 25, 2025



1 Introduction

The objective of performing the Charpy test is to experimentally determine the material's toughness and understand the behaviour of brittle and ductile materials under high-strain-rate conditions.

1.1 Ductile and Brittle Materials

- A Brittle material fractures or breaks to relatively small or no extension when an external force is applied. They absorb very little energy when the Charpy test is performed and break with a clean cut.
- A Ductile material can undergo a large strain or exhibit plastic deformation when an external force is applied before it fractures or breaks. They absorb much more energy than Brittle materials when the Charpy test is performed and break with a dull cut with jagged edges.

Table 1: Comparison between Ductile and Brittle Materials

Ductile Materials	Brittle Materials
Fail gradually by neck formation under external tensile loading.	Fail by sudden fracture without warning (no necking).
Show longer life when subjected to fatigue loading.	Fail faster when subjected to fatigue loading.

gh

1.2 Effect of Temperature on Material Behaviour

- Materials with body-centred cubic structure (BCC) at elevated temperatures become more ductile, meaning the ductility for these materials increases with an increase in temperature. In contrast, these materials exhibit brittle characteristics at low temperatures, signifying that brittleness increases with decreasing temperatures.
- The effect of temperature is so significant that at a specific temperature, Ductile materials start behaving like Brittle materials, and this transition is known as “Ductile-to-Brittle Transition”. This property of materials makes them susceptible to failure in regions of low temperatures.
- Materials with face-centred cubic structures (FCC), on the other hand, do not show different behaviours at different temperatures or undergo a ductile-to-brittle transition.

1.3 Effect of Strain Rate on Material Behaviour

- The ductility or brittleness of a material also depends on the strain rate with which it is loaded. At higher strain rates (fast loading), materials tend to behave more brittle because they don't get enough time to deform plastically. Whereas in the case of lower strain rates (slow loading), they have sufficient time to exhibit plastic deformation and hence tend to behave a little more ductile.
- In the Charpy V-notch test, since there is a very high strain rate due to the sudden impact of the hammer, the materials show a little higher brittle behaviour.

1.4 Toughness of Materials

- The toughness of a material is defined as the ability of a material to absorb energy and plastically deform before fracture takes place. It combines strength and ductility, meaning a tough material can withstand high stress and deform significantly before breaking.
- Toughness is often measured by computing the area under the stress-strain curve in a material's tensile test, representing the total energy the material can absorb before failure.



1.5 Difference Between Strength and Toughness

- Strength refers to the ability of a material to withstand a force without breaking or deforming permanently. Toughness differs from strength as it measures how much energy the material has absorbed during deformation before it fractures.
- A material can have high strength but low toughness, such as some brittle materials like cast iron, or it can have high toughness but lower strength, like some ductile metals.

2 Procedure

1. Switch on the power supply and remove the emergency stop of the machine.
2. Press the PENDULUM RETURN button to release the pendulum hammer and let it come to rest at the lowest point.
3. After the pendulum hammer has come to rest, press the ZERO button to set the current angle to 0 degrees.
4. Press the PENDULUM RAISE button to raise the pendulum hammer to its original position and ensure that the current angle displayed is -150° .
5. Dry run by pressing the RUN BUTTON to find the offset Energy Absorbed value by the machine.
6. Choose one of the samples from the three materials (Aluminium, Cast Iron and Mild Steel).
7. Use the support to place the sample on the anvil with the flat side facing the pendulum hammer, and the notch is aligned with the extension in the support.
8. Remove the support without disturbing the sample and ensure no one is standing near the machine or in the direction of the pendulum swinging.
9. Press the RUN Button to release the pendulum hammer and measure the readings of the absorbed energy on the display.
10. Re-run the experiments with different samples.

3 Observations

The readings of "Energy Absorbed" (in Joules J) obtained from the Charpy V-notch Test Machine for each material in multiple trials are listed below (Table 2), along with the corresponding calculated toughness (in MJ/m^3) values.

Student	Sample	Energy Absorbed (in J)			Toughness (in MJ/m^3)		
		Aluminium	Mild Steel	Cast Iron	Aluminium	Mild Steel	Cast Iron
Student 1	Sample 1	12.61	38.92	1.15	2.03	6.42	0.26
	Sample 2	10.16	34.32	1.36	1.62	5.65	0.29
Student 2	Sample 1	8.67	34.76	0.98	1.375	5.72	0.23
	Sample 2	10.07	22.37	1.02	1.61	3.66	0.24
Student 3	Sample 1	11.50	33.44	1.67	1.85	5.50	0.35
	Sample 2	10.16	40.35	1.41	1.62	6.655	0.305

Table 2: Obtained Energy Absorbed and Calculated Toughness

Calculation: The machine had an offset value (when no specimen placed) of 0.42 J. Hence, we subtracted the offset value from the readings recorded. Finally, to get the toughness of the material, we divide by the volume of the material where the dimensions of each sample is $10 \text{ mm} \times 10 \text{ mm} \times 6 \text{ cm}$ (neglecting the removed volume of the notch). A sample calculation is also done below (For Sample 1 of Student 1 for Aluminum).

$$\begin{aligned}\text{Actual Energy Absorbed} &= \text{Energy Absorbed Reading} - \text{Average Offset} \\ &= 12.61 - 0.42 \text{ J} = 12.19 \text{ J}\end{aligned}$$

$$\begin{aligned}\text{Toughness} &= \frac{\text{Actual Energy Absorbed}}{\text{Volume of Sample}} \\ &= \frac{12.19}{0.000006} \approx 2.03 \text{ MJ/m}^3\end{aligned}$$

Therefore the average toughness of the materials are:

- Aluminium = 1.68 MJ/m³
- Mild Steel = 5.60 MJ/m³
- Cast Iron = 0.28 MJ/m³

From the test and data, it was observed that Mild Steel showed the highest toughness, followed by Aluminium, while Cast Iron showed the lowest toughness. It was also observed that Aluminum and Mild Steel showed higher plastic deformation on impact of the hammer than Cast Iron before fracture.

It was also observed during the test that the cast-iron specimen, being more brittle, broke into two pieces on impact, with a nearly flat plane through the notch. While mild steel and aluminum being more ductile, most of the specimen plastically deformed into V-shape and not completely broke to pieces. A few pieces that broke were highly deformed at the notch, and the plane of cut wasn't flat either. The images for each specimen after impact are shown below.



(a) Mild Steel



(b) Cast Iron



(c) Aluminium

4 Conclusion

From the Charpy V-notch test, it can be concluded that usually ductile materials have higher toughness as compared to brittle materials at the same temperature. The experiment also helps to conclude the fracture behaviour of different materials. Ductile materials absorbed a higher amount of energy and showed significant plastic deformation, whereas brittle materials absorbed little energy and fractured suddenly with very little deformation.

5 References and Resources

1. <https://www.sciencedirect.com/topics/materials-science/ductile-material>
2. <https://www.sciencedirect.com/topics/materials-science/ductile-to-brittle-transition>
3. <https://www.difference.minaprem.com/solid/difference-between-ductile-material-and-brittle-material/>
4. <https://www.samaterials.com/content/toughness,-hardness,-and-strength.html>